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A Statistical Audit of LLM Calibration Heterogeneity Across Knowledge Domains

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└ What is calibration — and why does it matter?

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A model is **well-calibrated** if its stated confidence tracks empirical accuracy:

- 80% confident $\Rightarrow$ correct 80% of the time

Miscalibration is typically reported as a **single number** (Expected Calibration Error):

$$\text{ECE} = \sum_b \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|$$

The problem: aggregate ECE over 57 subjects, 4 domains, 14,042 questions hides where and how much miscalibration occurs.

Analogy: reporting one accuracy number across demographic groups is insufficient for a fairness audit. The same logic applies to calibration.

[45 sec]

Open with the definition — make sure the audience knows what calibration means before anything else. The formula is on the slide so you don't need to dwell on it; just say “ECE is the weighted average gap between confidence and accuracy across confidence bins.”

The key move is the fairness analogy: just like we'd never accept a single accuracy number as a fairness audit, a single ECE number is not a calibration audit. Land that point before moving on — it's the whole motivation.

If anyone in the room works on ML deployment, you can add: “when a model says it's 90% confident, users treat that as a reliability signal — if that signal is only accurate on average, users in certain domains are being systematically misled.”

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└ Research questions

Research questions

- ❖ How much of calibration variance is attributable to domain vs. subject vs. individual question?
- ❖ What question features predict the direction and magnitude of miscalibration?
- ❖ Are patterns consistent across models, or model-specific?

Framework

Three-level mixed model + isotonic calibration curves + empirical Bayes shrinkage + Benjamini-Hochberg FDR correction

[30 sec]

Read the three questions quickly — the audience will follow. For the framework block: “I won’t spend time on each method now; the key idea is that we want statistically rigorous answers, not just descriptive bar charts.”

Don’t explain what BH correction is here — that comes up naturally in the robustness slide. Just name-drop it so the audience knows it’s coming.

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└ Data & confidence elicitation

Data & confidence elicitation

MMLU: 14,042 four-choice questions, 57 subjects, 4 domains (questions → subjects → domains)

Two models:

- GPT-4o-mini (OpenAI API, top_logprobs=20)
- Llama-3.8B-Instruct (Colab T4, 4-bit quantization)

Confidence measure — renormalized softmax over the four answer tokens:

$$c_i = \max_{a \in \{A, B, C, D\}} \frac{e^{q_a}}{\sum_{j \in \{A, B, C, D\}} e^{q_j}}$$

Primary outcome — signed calibration gap:

$$\text{gap}_i = c_i - y_i \quad (y_i \in \{0, 1\})$$

Positive = overconfident; negative = underconfident.

[45 sec]

Explain the hierarchy briefly: “MMLU has 57 subjects like clinical medicine, formal logic, high school math — grouped into 4 broad domains. That nested structure is what makes a multilevel model the right tool.”

For the confidence measure: “We extract the log probability the model assigns to each of the four answer tokens, renormalize them to sum to 1, and take the max. This is a standard approach — it gives us a proper probability on the model’s chosen answer.”

For the signed gap: “The nice thing about using the signed gap instead of ECE is that it tells us the *direction* — positive means the model thinks it knows more than it does.”

One methodological note if it comes up: we had to use top_logprobs=20 for GPT because with 5 logprobs, 55% of questions had at least one answer token missing, which would artificially inflate confidence estimates. That’s a real data quality issue that took some debugging.

[If asked: what exactly is the signed calibration gap?] The gap is computed at the *question level*: for each of the 14,042 questions, we take the model’s renormalized confidence $c_i \in (0, 1]$ (the max softmax probability over A/B/C/D) and subtract the binary correctness indicator $y_i \in \{0, 1\}$. So $\text{gap}_i = c_i - y_i$ is a number in $(-1, 1)$ for every individual question. It is positive

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└ Hierarchical model

Hierarchical model

Three-level mixed model

Questions (i) nested in subjects (j) nested in domains (k):

$$\text{BAP}_{ijk} = \mu + \mathbf{x}_{ijk}^T \beta + u_k + v_j + \varepsilon_{ijk}$$

Fixed effects

μ : global intercept (mean overconfidence)
 β : question-level covariates
 (word count, entropy, negation,
 max choice length) — all standardized

Random effects

$u_k \sim \mathcal{N}(0, \sigma_{\text{domain}}^2)$
 $v_j \sim \mathcal{N}(0, \sigma_{\text{subject}}^2)$
 $\varepsilon_{ijk} \sim \mathcal{N}(0, \sigma^2)$
 Estimated via REML
 (statamodels.REML2X)

ICC decomposition: what fraction of calibration gap variance is between-subject vs. within-subject?

[30 sec]

Walk through the model briefly: “We model the signed calibration gap — confidence minus accuracy — at the question level, with random intercepts for both subject and domain.”

Emphasize the key question: “The ICC decomposition will tell us whether the miscalibration is driven by specific subjects, or whether it’s spread evenly across all questions regardless of subject.”

Mention REML: “We use restricted maximum likelihood, which gives unbiased variance component estimates — important here since the variance decomposition is our main inferential target.”

[If asked: why fixed effects for questions and random effects for subject/domain?] The core justification is about whether you have a theory worth committing to.

At the **question level**, you have interpretable, measurable features and a plausible theory: longer questions are harder to reason about; higher entropy means the model is more uncertain; negations are known to trip up models. You can name the features, measure them, and interpret the coefficients. Fixed effects are the right tool when you have a theory.

At the **subject and domain level**, you don’t have a good theory of *which* subject-level features

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— Finding 1: heterogeneity is real but concentrated at the question level

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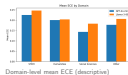
Table: Variance decomposition (both models)

Level	GPT-4o-mini		Llama-3.8B	
	Var	ICC	Var	ICC
Domain	0.000	0.000	0.000	0.000
Subject	0.013	0.079	0.004	0.020
Residual	0.150	0.922	0.190	0.980

Both models: between-subject structure exists but is modest.

GPT: 7.9% between-subject ICC.

Llama: 2.0% — more diffuse miscalibration.



[30 sec]

Lead with the structure: “Both models show that the vast majority of calibration variance — 92 to 98% — is at the question level. But a meaningful slice is systematically concentrated at the subject level. That’s not noise — it’s structure.”

Highlight the difference: “GPT shows 7.9% between-subject ICC, Llama shows 2.0%. Llama’s miscalibration is more diffuse — spread more evenly across subjects rather than concentrated in particular domains.”

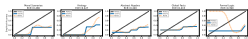
[Important: what ICC does and does not say] ICC = 7.9% does NOT mean calibration barely varies across subjects. It means 7.9% of total question-level variance is attributable to subject membership. That 7.9% is still large enough to produce a 13× range in mean subject gaps (0.037 to 0.487) — the two facts are compatible. Most question-to-question variability is noise regardless of subject (the 92%), but the systematic component that does exist is substantively meaningful.

Be upfront about domain ICC = 0: “Domain-level variance is estimated at zero. That doesn’t mean domains don’t matter descriptively — we see clear differences in the bar chart. But the model can’t separate domain-level variance from subject-level variance because the subject

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└ Finding 2: all 57 subjects are overconfident on average

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The 5 most miscalibrated subjects shown. All 57 subjects have positive mean calibration gap. In the high-confidence region where data is concentrated, curves lie below the diagonal throughout. Isotonic (solid) and kernel-smoothed (dashed) curves agree where support exists; divergence at low confidence reflects empty bins.

[30 sec]

Lead with the universal claim carefully: “All 57 subjects have a positive mean calibration gap — they are overconfident on average. These 5 curves show what that looks like in the region where data is actually concentrated, which for most subjects is the high-confidence end.”

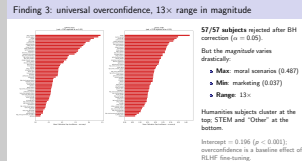
Be precise: the below-diagonal pattern holds in the region of support (roughly 0.85–1.0 for most subjects). It is NOT correct to say every point on every curve is below the diagonal — 23/57 subjects have parts of their isotonic curve above the diagonal, but this is at low-confidence bins where there are essentially no data points. The mean gap being positive is the correct universal claim.

On the formal_logic kernel curve specifically: the dashed curve reads near 1.0 from confidence 0.03 to 0.53. This is a smoothing artifact — formal_logic has zero questions in those bins (126 total questions, 89 of which are in the 0.95–1.00 bin). The kernel is smoothing over empty space and should not be interpreted there. The isotonic curve is more honest: it just holds the last observed value flat across empty bins.

Point out isotonic vs. kernel agreement where it matters: “In the region where data exists — above 0.7 or so — the solid and dashed lines track each other, which validates the monotonicity

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— Finding 3: universal overconfidence, 13× range in magnitude



[45 sec]

Preempt the obvious question: “57/57 rejected sounds impressive, but it’s largely a consequence of large sample size and strong global overconfidence — not a deep finding. The interesting result is the 13-fold range in magnitude.”

What is the 13× Simply the ratio of the largest to smallest mean gap: moral scenarios (0.487) divided by marketing (0.037) $\approx 13\times$. The model is 13 times more overconfident on moral scenarios than on marketing. That heterogeneity is invisible in any single aggregate ECE number.

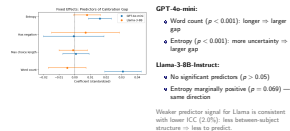
“A gap of 0.487 means the model is, on average, nearly 50 percentage points more confident than it is accurate. A gap of 0.037 means it is essentially well-calibrated. Those are qualitatively different deployment situations.”

[If asked: why is 57/57 not a meaningful finding?] Intuition: imagine testing whether a coin is slightly biased. With 10 flips you might not detect it. With 1,000,000 flips you would detect even a 50.001% bias as “statistically significant” — but that doesn’t mean it matters. Same here: with 14,042 questions and a model already globally overconfident by 0.196 on average, you have enough power to flag even a trivially small true effect. Even marketing (gap = 0.037

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What predicts the gap? Fixed effects

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[30 sec]

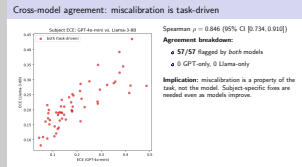
Lead with the GPT entropy finding — it's counterintuitive: "When GPT-4o-mini spreads its probability mass across options, expressing uncertainty, the gap actually gets larger. The model isn't reliably correct more often when it's hedging; it's miscalibrated even in its expression of uncertainty."

Then contrast with Llama: "Llama shows no significant predictors. The trend for entropy is in the same direction but doesn't reach significance. This is consistent with Llama having less between-subject structure overall — there's simply less systematic variance to explain."

[If asked: why do GPT and Llama differ in fixed effect significance?] GPT's ICC is 7.9% — its miscalibration is organized by subject, concentrating in subjects like abstract algebra and moral scenarios. These subjects also tend to have longer, more ambiguous questions. So word count and entropy are picking up real signal because they correlate with the hard subjects. Llama's ICC is only 2.0% — its miscalibration is more diffuse, not concentrated in particular subjects. There is no such structure for question features to exploit. On top of that, Llama's residual variance is larger in absolute terms ($\sigma_\epsilon^2 = 0.190$ vs. 0.150 for GPT), so the same fixed effects face more noise, inflating standard errors and making significance harder to reach.

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└ Cross-model agreement: miscalibration is task-driven



[45 sec]

[Big picture: how this slide connects to earlier findings] Three separate claims have been made so far — it is easy to conflate them:

1. **Universal overconfidence** (global intercept = 0.196): all 57 subjects are overconfident because the model itself is globally biased by RLHF. The 57/57 BH rejection is mostly detecting this global effect, not between-subject differences.
2. **Between-subject heterogeneity in magnitude** (ICC = 7.9%, 13 $\times$ range): on top of the global bias, subjects vary in how badly miscalibrated they are. 7.9% of total variance is between-subject — modest, but enough to produce a 13 $\times$ range in mean gaps that matters practically.
3. **Task-driven miscalibration** (this slide, $\rho = 0.846$): whatever between-subject variation exists is consistent across two very different models. This answers a different question than ICC — not “how much variation is there?” but “where does that variation come from?”

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└ Sensitivity analysis: are findings robust?

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Three stress tests — run on both models.

1. ECE binning & NLCS

$b \in \{10, 15, 20, 25, 30\}$ bins.

GPT: $\rho \geq 0.997$

Llama: $\rho \geq 0.984$

vs. unbinned NLCS: $\rho \geq 0.861$

Rankings stable.

2. FDR threshold

BH at $\alpha \in \{0.01, 0.05, 0.10\}$.

GPT: 56/57/57 rejected

Llama: 55/57/57 rejected

Not driven by α .

3. Model structure

3-level vs. 2-level model.

GPT: $ICC_{adj} = 0.079$

Llama: $ICC_{adj} = 0.020$

(identical in both specs)

ICC robust.

[45 sec]

Frame briefly: “Before trusting these results, I stress-tested the three main modeling choices across both models.”

Column 1: “Subject rankings don’t move when you change the number of ECE bins, and they correlate strongly ($\rho \geq 0.861$) with the unbinned Negative Log Calibration Score (NLCS). Our findings aren’t artifacts of binning choices.”

Column 2: “At $\alpha=0.01$, GPT still rejects 56/57 and Llama rejects 55/57. The two Llama subjects that drop out at the strictest threshold have adjusted p-values of 0.014 and 0.031 — just above the cutoff. The finding is not alpha-sensitive.”

[If asked: what does varying the BH α tell us?] BH α controls the expected proportion of false discoveries among all rejected tests. At $\alpha = 0.05$, at most 5% of rejections are expected to be false; at $\alpha = 0.01$, at most 1%.

If rejections were borderline — p-values clustered just below the threshold — tightening α would flip many decisions. The fact that results barely move (57/57 $\rightarrow$ 56/57 for GPT, 57/57 $\rightarrow$ 55/57 for Llama) shows almost all subjects have adjusted p-values so far below any reasonable threshold that the BH correction barely matters. The two or three subjects that drop at $\alpha = 0.01$

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Takeaways

Takeaways

- ❖ **The unit of analysis is the subject, not the model.** Aggregate ECE hides a $13\times$ range in miscalibration magnitude. 57/57 subjects rejected is the wrong headline — the magnitude variation is what matters.
 - ❖ **Miscalibration is task-driven, so subject-level fixes are needed.** Spearman $\rho = 0.846$ across two very different models means the same subjects are hardest regardless of architecture or training. Global temperature scaling cannot fix subject-specific structure.
 - ❖ **The audit framework generalizes.** Multilevel model + BH correction + shrinkage applies to any benchmark with a hierarchical structure — a reusable tool for honest calibration evaluation.
 - ❖ **Open questions remain.** Two models is suggestive, not definitive. We can describe which subjects are hard but not why — abstractness, training data rarity, or something else is still unknown.
- Next steps: subject-specific temperature scaling; broader cross-model comparison; extend to open-ended generation.

[45 sec]

Don't read the bullets — synthesize across all four:

“The core finding is that the subject is the right unit of analysis for calibration, not the model. Aggregate ECE hides a 13-fold range; 57/57 rejected sounds impressive but is mostly a consequence of large sample size and global overconfidence — the magnitude variation is the finding that matters.”

“The reason subject-level recalibration is needed, not model-level, is that $\rho = 0.846$ across two very different models tells us the hardness is in the knowledge domain itself. You cannot train your way out of it.”

“We're honest about what we don't know: two models is suggestive not definitive, and we can describe which subjects are hard but not why. That's the natural next step.”

Land the practical implication last: “If you are deploying an instruction-tuned model on professional tasks — legal, medical, mathematical — the aggregate calibration number is not what you should be reporting. The gap is 13 times worse on some subjects than others, and that pattern holds across models.”

End confidently: “The framework is the contribution. It gives you a statistically honest answer